TRUNKBELAY · FEASIBILITY ASSESSMENT

TrunkBelay feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/trunkbelay		
Source	docs/08-feasibility.md @ 7aa480e		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	O8 decided (2026-10-25)
0.1	2026-10-25	Amish Chadha	First assessment (kit 1.9.1 upgrade) of the TRL 3 design, option A

TrunkBelay feasibility assessment

Feasible on paper, with one deciding unknown: the collar (option A) is constructable from plate, tube and bought rigging, its lowering numbers hold with margin, and its parts are easy to buy, but whether it locks on a wet palm depends on a friction nobody has measured, and its strength margin is below the low end of rescue practice in the worst case, a safety trade-off Amish decided on 2026-10-25: a 3.5 kN proof load and a test to failure of a sample frame at TRL 4, with a stronger frame as the fallback (O7). Method: STANDARDS section 23.

> **Safety:** Feasible on paper is not safe to use. Nobody is lowered with this kit before the collar has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design: grip on the trunk	The self-locking couple is sound physics; the full equilibrium gives a locking factor of 1.63 against 1.54 in the simplified method, both on an assumed combined friction of 0.9; settlement 14 to 18 mm with pad shear against 20	Medium	Friction of rubber and sleeved chain on wet coconut bark is unmeasured; if it were below 0.58 the collar would slide on a 200 mm trunk	TKB-STM-001 S3, S4; TKB-CAL-001 [B1] to [B5]
Design: strength	Every frame part at 1.21 times its welded proof strength or better at 2.5 kN, even with all of the chain's pull on one slot (D-S2); chain 4.1 in one leg, rope 5.7 at the knot	Low	Estimated breaking load about 5.0 kN, 5.1:1 on the one-person load, 4.5:1 in the worst chain split: below the 5:1 to 10:1 of rescue practice (TKB-STM-001 S30, plausible until the TRL 4 test to failure; TKB-DEC-001 O7, decided 2026-10-25)	TKB-STM-001 S6 to S12, S30; TKB-CAL-001 [C1] to [C6], [C1b], [C4b]
Design: lowering	80 N at the brake hand; 25 m lowering turns 24.8 kJ into heat, within a class A descender's rating	High	The bought descender's own data replaces the assumed rope friction	TKB-STM-001 S13, S14, S29; TKB-CAL-001 [D1], [D2]
Design: use at height	Rigging about 4.5 minutes against 5; collar must be set at least 300 mm above the victim's attachment; on a leaning trunk the lock falls toward 1.0 at 30 deg	Medium	Thin time margin; whether a rescuer can reach above the attachment is only shown by a drill; 20 deg lean limit and frame-side drill rule decided by Amish on 2026-10-25 (O8)	TKB-STM-001 S17, S24, S31; PR-002, PR-004
Materials	6082-T6 plate and RHS, natural rubber, grade 80 chain, polyamide rope and webbing	Medium	Galvanic contact of steel and aluminium, polymer ageing in sun and heat (plausible, not shown)	TKB-STM-001 S25, S26
Manufacturing and assembly	Profile cutting, drilling, AC TIG welding, rubber bonding, rivet nuts; ten assembly steps	Medium	An AC TIG welder and a welder who knows aluminium are needed; weld quality decides the frame's strength	TKB-BLD-001; TKB-DWG-101 to 109
Power and controls	None: a passive mechanism worked by hand	High	Not applicable	TKB-PRC-001
Supply of parts	Chain, carabiners, descender, rope, triangle and slings are catalogue items	High	Certified rescue gear is most of the cost (USD 280 descender, USD 140 triangle in the BOM; a cited branded descender lists at USD 421)	bom/bom.csv; TKB-MKT-001
Model and checks	Detailed D3 model of every option passes the fidelity check; 244 constructability checks pass on 200, 300 and 450 mm trunks	High	The D3 model found one fit failure (tether quick link, D-S1) and the independent review one strength failure (cleat web, D-S2), both fixed	cad/fidelity.json; `python cad/src/model.py --check`

Ratings: High, Medium, Low.